README

Post-bop modal-tonal AI jazz improvisation

Program(me) Notes

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WOLFSON
COLLEGE

Now’s the time

- Back in the late 1900s I was part of a jazz workshop helping people learn to play jazz.
- One of the exercises was called "slick licks" – we would sit in a circle and “trade licks” on our instruments.
- A big barrier for jazz learners can be finding people to play with.
- Q. Can I build a software system to enable people to practice their slick licks? And specifically… can a bass player use this?!
- It should listen and respond, sometimes lead, be realistic enough to be useful.
- NB This interactive “robot” is designed for practice purposes, and it's not intended to replace a human player. Human interaction was key to slick licks!

Program(me) Note

*README** is a live conversation between a human bassist and an artificial musician that listens, responds, and remembers. Wolfson is a computer system trained on hundreds of jazz saxophone solos; it hears each bass phrase, analyses its character – the notes, the rhythm, the energy – and improvises a reply in real time. Over five minutes the music traces a journey through jazz history: an exploratory, chromatic opening gives way to the modal sound of Miles Davis and Herbie Hancock, builds to a hard-driving peak, and settles into a meditative close. No two performances are identical.

- The title works on three levels: the *README* file is the first thing you read when you encounter a piece of software; the software reads me, the bassist, responding to whatever I play; and it is a sideways glance at *Recorda Me* ("remember me”) a jazz standard, and a reminder that all improvisation is, in some sense, an act of memory.

All the things I have been* thinking about

Phrase analysis
Question and answer
Leadership & role swapping
Opening echo
Contour steering
Chord conditioning
Harmonic modes
Scale pitch bias
Pitch range
Swing / triplet feel
Live tempo tracking
Dynamics

Generation temperature
Phrase-shape dynamics
Articulation
Bass pitch-class tracking
Energy arc
Motivic development
Rhythmic displacement
Lyrical motif re-use
Voice leading
Modal leap bonus
Stochastic thinning
Phrase breathing
Singable duration bias

Rhythmic density
Rhythmic complementarity
Register contrast
Phrase length control
Beat-matching / trading bars
Repetition control
Sax riff / insistence
Proactive mode
Live phrase peeking
Beat-synchronised entry
Turn-taking
Loop mode
Arc timing

** Good name for a song?*

Fly me to the moon

The five giant steps in the arc…

Sparse	Exploratory, establish motifs slowly
Building	Groove warming up, modal
Peak	Fast, active, max rhythmic intensity
Recapitulation	Lyrical return, motifs recalled
Resolution	Slowest, long tones, fading out

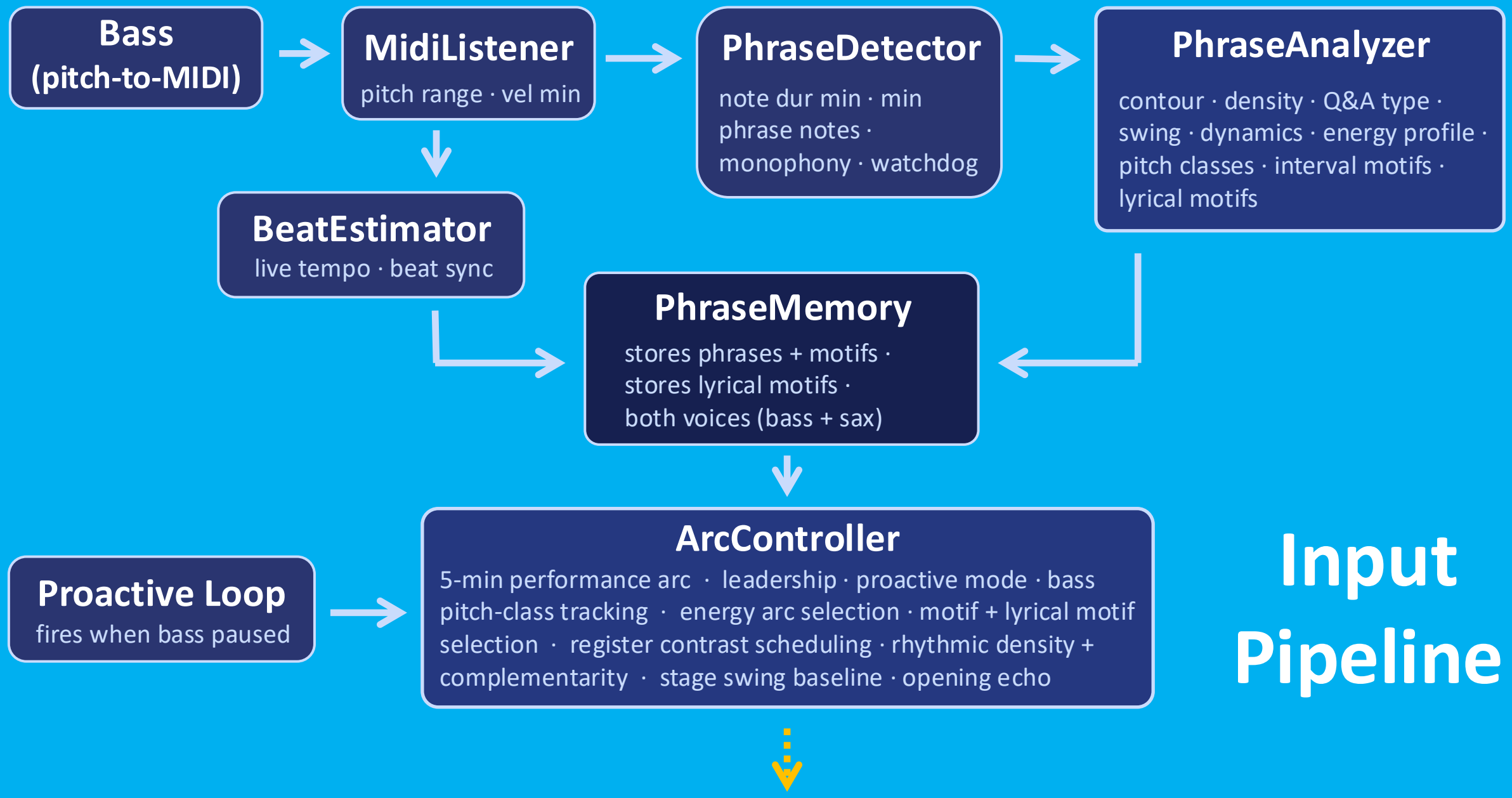
Sentimental journeys

This is inspired by:

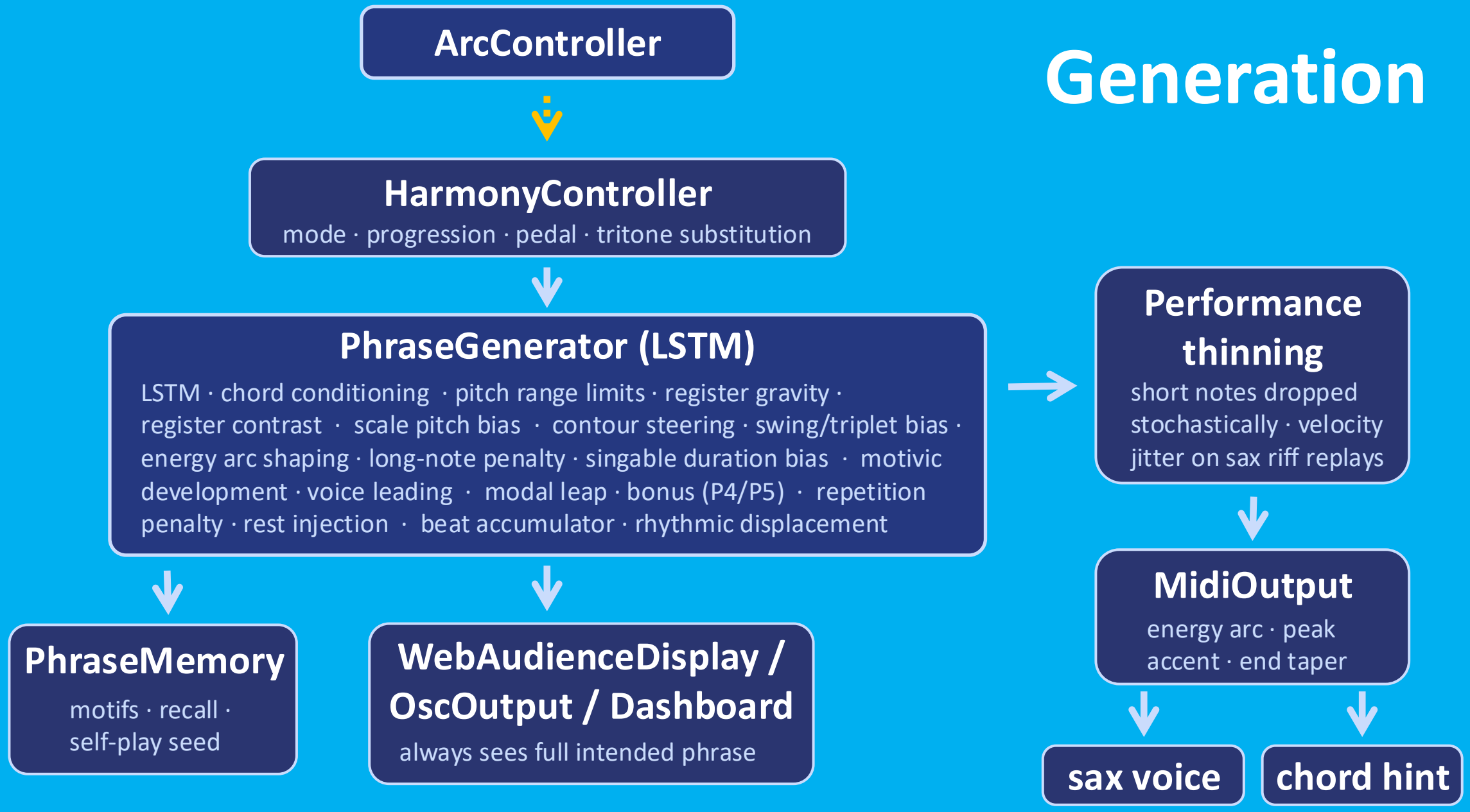
- *Post-bop modal-tonal hybrids* like:
  - **Milestones** (Miles Davis), **Little Sunflower** (Freddie Hubbard), **Maiden Voyage** (Herbie Hancock), **Footprints** (Wayne Shorter), and the atmosphere of **In a Silent Way** (Miles Davis). These songs are my favourite things*.
- The arc also traces a journey through jazz history :
  - Sparse opening resembles the chromatic freedom of free jazz,
  - Modal building stage reflects the Miles/Coltrane modal period,
  - Peak combines hard bop drive with modal intensity,
  - Pedal resolution has the meditative quality of late Coltrane e.g. A Love Supreme.

* *John Coltrane*

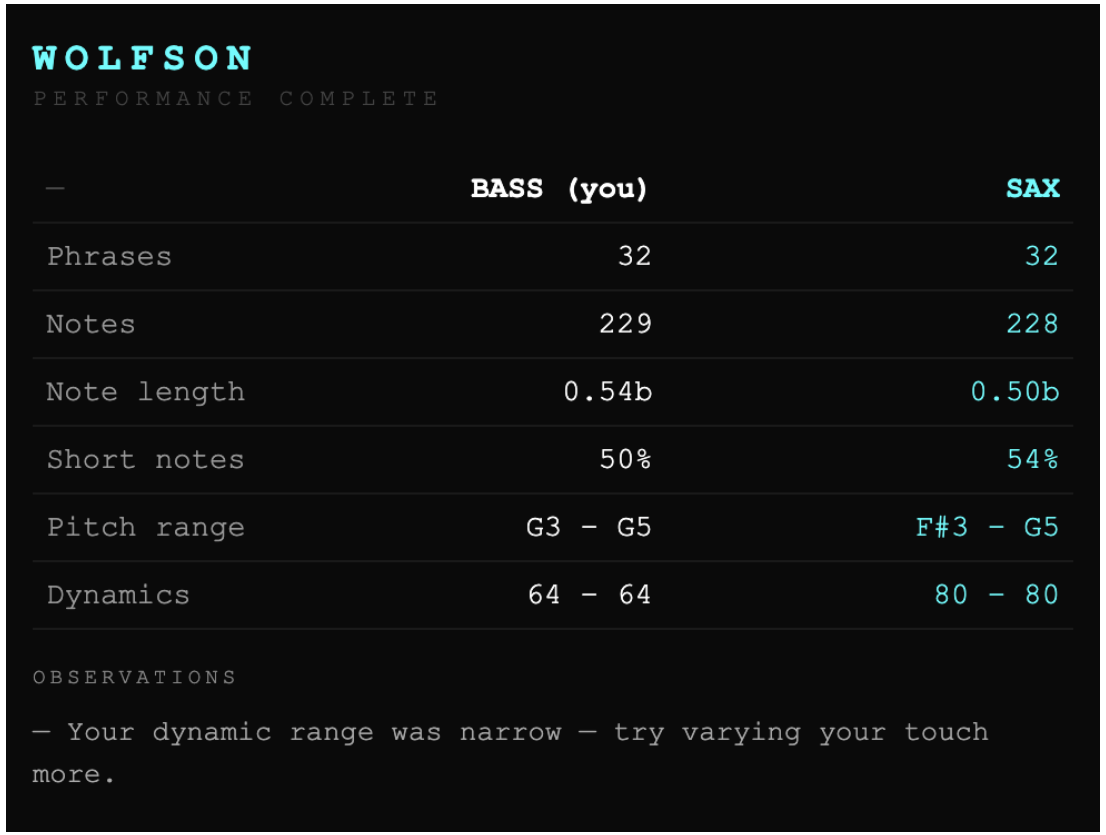
Input Pipeline



Generation


ArcController
HarmonyController
mode · progression · pedal · tritone substitution
PhraseGenerator (LSTM)
LSTM · chord conditioning · pitch range limits · register gravity · register contrast · scale pitch bias · contour steering · swing/triplet bias · energy arc shaping · long-note penalty · singable duration bias · motivic development · voice leading · modal leap · bonus (P4/P5) · repetition penalty · rest injection · beat accumulator · rhythmic displacement
PhraseMemory
motifs · recall · self-play seed
WebAudienceDisplay / OscOutput / Dashboard
always sees full intended phrase
Performance thinning
short notes dropped stochastically · velocity jitter on sax riff replays
MidiOutput
energy arc · peak accent · end taper
sax voice
chord hint

Soul A-eyes



Watch live during the performance on

wolfson.numbersintonotes.net

- Repo is on github.com/davidderoure/wolfson

- Acknowledgement:
  - Ray d'Inverno (for jazz, Lisp, AI, slick licks)
  - Mark d'Inverno (for jazz, AI, creativity, slick licks)
  - George E Lewis (for Voyager and Forager)
  - Robert Laidlow (for Alter-ing Ada Lovelace, and sax)
  - Hongshuo Fan (for Music Gesture Recognition)
  - Bofan Ma (for always enabling musicality)
  - Maria Kallionpää (for Climb!)
  - Ada Lovelace & Emily Howard (for Numbers into Notes)
  - Wolfson Digital Research Cluster, DiSc & TORCH (for AIMC 2024)

- *Grand Challenges for Computing Research* – Tony Hoare

Radar of the lost archive

- "The Grand Challenges Exercise of the UK Computing Research Committee"
- A workshop held in Edinburgh in November 2002, considered 109 submissions.
- Archival research suggests my submission was "The creative electronic musician". Or maybe "Interactive Electronic Musician".
- Sir Tony Hoare, Wolfson Emeritus Fellow, was Chair of the programme committee.

National e-Science Centre

Workshop on Grand Challenges for Computing Research

Sponsored by the UK Computing Research Committee, with support from EPSRC and NeSC

UK COMPUTING RESEARCH COMMITTEE

Panel C Submissions

5. Deep Adaption
6. Virtual Realms
7. If we could see fit
8. The creative electronic musician
9. Getting Androids to learn from testimony
10. Rendering with intent
11. Deep Thought

Someday my footprints will come but we might not be quite there yet:

- It uses MIDI: no pitch inflection, vibrato, or continuous dynamics within notes – every note is a rectangle.
- Time feel is mechanical within phrases: no microtiming, no laying back or pushing the beat.
- Conversation is polite by design – complementary, never confrontational, doesn't interrupt, crowd the register, or refuse to listen..
- It's pretty much in 4/4, no waltzes or take 5 – the tempo estimator assumes equal beats.
- No personal voice – an amalgam across an era rather than any one player's sound.
- The arc is clock-driven, not music-driven – deliberately for this performance (but see *Climb!*)
- Motivic development might feel like repetition – because it is.
- Trained on solos not melodies – no quoting, no playing off the tune.
- Trained on one era and school – the bebop/hard-bop vocabulary of the Weimar Jazz Database; no awareness of what came before or after.
- Inside/outside distinction and narrative arc within a phrase could be designed more explicitly – the model has learned some of this from transcriptions.
- It only listens to one instrument (but see *Forager)*.
- Audiences may find fixed, familiar harmony more immediately satisfying – responsiveness to the bassist means harmony is earned rather than given.

Wolfson — Performance Plan

Bass improvisation cues by arc stage — 1 May 2026, Wolfson College Oxford

Sparse 0:00–1:00	Building 1:00–2:30	Peak 2:30–3:30	Recapitulation 3:30–4:30	Resolution 4:30–5:00
• **Phrases echo back literally** — same intervals and rhythm, transposed to sax register. Play short, distinctive phrases. • **Leave 2+ beats of rest** after each phrase — system needs ~2 s silence to close the phrase and respond. • **Play quietly** — sax mirrors your dynamic. A quiet opening makes later contrast dramatic. • **Plant a 2–3 note motif** — it will be recalled in recapitulation.	• **Play D Dorian** (D E F G A B C) — 4+ distinct pitch classes shifts the sax harmonic language. Switch away briefly to hear it change. • **Repeat a riff 3+ times** — sax enters development mode: stronger motivic use, directed contour. It starts to *argue with* the riff. • **Play continuously (~8 s)** to trigger trading mode — sax breaks in, then re-enters every ~4 s (just under a bar). • **Let the sax repeat** (`--sax-riff-prob`) — sax insists on its phrase; you develop underneath. After two replays it varies itself.	• **Play loudly and densely, then sparsely** — busy bass invites sparser sax; a terse phrase invites a rush of notes. • **Rising phrase then falling** — contour steering. Sax responds to a rising line with a descending resolution and vice versa. • **Switch between Q & A and continuous playing** — short phrases with 2-beat rests give call-and-response; sustained riff triggers trading mode. The contrast is most audible here.	• **Play low** — register contrast pushes sax into upper register. Spatial separation is most audible here. • **Listen** — sax quotes motifs from sparse and building stages. Pause to let the audience hear the system remembering.	• **Stop playing** — sax closes the piece unaccompanied. Put the bass down deliberately. The clearest demonstration of musical agency.

Q & A mode: play a phrase, stop for 2+ beats — sax replies ~1 s after silence. **Trading mode:** play continuously for ~8 s — sax breaks in and re-enters every ~4 s. **Echo:** first 4 exchanges are literal echoes; after that Wolfson generates freely. **Motif:** repeat sparse motif 2–3× for strong recapitulation recall. **Auto-start:** Wolfson plays first; join when ready.